Tanuj Ranjith

High School Student - Robotics, Computer Vision & Embedded Systems

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SUMMARY

High school senior with hands-on experience in robotics, AI, and software engineering. Led electronics design and drivetrain for a VEX team that reached Worlds. Contributed to the Open-Source Leg prosthesis project. Provisional Patent Applicant (64/111,756). Founded VetsITGuide, an AI career platform for veterans with 500+ users. Proficient with Autodesk suite, Python, C++, AI agents, and Vision-Language Models (VLMs). Recognized through multiple university programs and courses from Harvard, Stanford, Duke, Michigan, Microsoft and Google Cloud.

EDUCATION

B. Reed Henderson High School, West Chester, PA *Senior (Class of 2027)*

- AP: Computer Science Principles, Computer Science A, AP Physics 1, AP Physics C, AP Cybersecurity, AP Calculus AB, AP Calculus BC, AP English Language and Composition, AP English Literature and Composition, AP U.S. Government & Politics, AP United States History, AP Psychology
- Honors: Biology, Pre-Calculus, Algebra 2, Engineering Design and Architecture 1-4
- GPA: 4.4 / 4.0

EXPERIENCES & INTERNSHIPS

Acuate IT Inc, West Chester, PA - Junior application developer (part-time) *Jul 2024 - Present*

- Built the Stoovi application and agentic AI workflows to automate business processes.
- Developed a shared Python framework for integrations, cutting integration issues by approximately 30%.

Open-Source Contributor - Neurobionics Lab, Open-Source Leg, University of Michigan *Sep 2025 - Apr 2026*

- Contributed to the open-source-leg prosthesis project by auditing, fixing, and testing installation tools and commands.
- Updated outdated links and refined setup documentation and Collaborated with university researchers.

AI/ML Intern - HUVTSP, Vanguard Defense El Segundo, CA (Remote) *Jun 2025 - Aug 2025*

- Trained a YOLO classification model to ~85% accuracy to classify 80+ military aircraft types.
- Led the aerial/satellite data track, ensuring high-quality labels for downstream object-detection models. (GitHub).

RESEARCHS & PATENTS

Predictive Cable Slack Compensator for Tendon-Driven Robotic Joints *Mar 2026 - Present*

Filed a U.S. provisional patent application (pro se) for a mechanism and control method that predicts and compensates for cable slack in tendon-driven robotic joints, improving precision and reliability in cable-actuated systems such as prosthetics and dexterous robotic manipulators; nonprovisional conversion in progress.

Independent Researcher (Co-Author) | Quantum Information & Error Correction *Nov 2025 - Present*

- Co-authored a quantum error-correction study and ran 4,000-shot Qiskit simulations. The five-qubit code reduced logical errors by 45% at a physical error rate of p = 0.10.
- Analyzed performance across bit-flip and phase-flip noise models to determine logical error rates. The paper was reviewed by Dr. Bryan Gard, Senior Scientist at Georgia Tech.

PROJECTS

VetsITGuide.com - AI Career Guide for Veterans, West Chester, PA *Jan 2024 - Present*

- Founded and built an AI-powered platform that helps veterans transition into IT and cybersecurity careers.
- Accepted into NVIDIA Inception program and secured $17,500.
- Recognized by US Department of Veterans Affairs and PA Senator Carolyn Comitta with 500+ registered users.

Sutra - Student operating system, West Chester, PA *Aug 2025 - Sep 2025*

- Built Sutra, a local-first student productivity platform combining notes, canvas pages, classes, assignments, habits, study tools, focus mode, and an AI assistant into one private browser-based workspace.
- Developed Sutra Assistant to help students organize coursework, generate study materials, plan tasks, and interact with their workspace using AI-powered academic context (GitHub).

Additional Projects

- Income Classification - Stanford Pre-Collegiate Program: Built a classification model using the Adult Income dataset (UCI ML Repository) to predict whether individuals earn over or under $50K annually. Trained and fine-tuned multiple models including Logistic Regression, Decision Tree, Random Forest, AdaBoost, Gradient Boosting, XGBoost, Bagging Classifier, and Support Vector Machine (SVM).
- iVoted.com - Brought voting awareness in the local community by building a website to emphasize the importance of voting and distributing door-to-door flyers.
- Pong with NEAT - Python game where an AI opponent is trained using the NEAT algorithm.
- Endangered Animals - Created a website using Python, JS, CSS to build awareness for Animal Endangerment.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

VEX Robotics - Lead Coder, Exton, PA *Aug 2020 - Present*

- Program autonomous routines using VEX Pros C++ with PID control, tracking wheels, encoders, and inertial sensor.
- Use odometry to build accurate autonomous paths for competition robots.

Technology Student Association (TSA) - Secretary, B. Reed Henderson High School *Aug 2025 - Present*

Participated in TSA Architectural Design, TSA Webmaster, and TSA Dragster Design. Led social media & outreach, and coordinate members while channeling feedback to leadership.

Artificial Intelligence (AI) Club - President, B. Reed Henderson High School *Aug 2025 - Present*

Led hackathon participations and assisted in leading the club. Collaborated closely with the President and other executive members, and supported the successful planning and execution of club activities. Oversaw projects and handled technical projects and membership engagements for the club.

National Honor Society - Member, B. Reed Henderson High School *Apr 2026 - Present*

Inducted April 2026 at the Henderson Auditorium.

SKILLS

- Programming: Python, Java, JavaScript, C++, Django, NumPy, Pandas, PyTorch, TensorFlow, HTML5, CSS
- CAD & Design: AutoCAD (2D/3D), Inventor, Revit
- Fabrication: Drill press, band saw, 3D printer, laser cutter, Dremel, chop saw, and related shop tools

CERTIFICATES & AWARDS

- **Certificates:** Harvard: CS50's Introduction to Programming with Python; CS50's Web Programming with Python and JavaScript, Stanford - Intro to Machine Learning (Pre-Collegiate Program), Duke - Data Science with NumPy, Sets, and Dictionaries; Advances in AI Workshop, University of Michigan's Joy of Coding, MedTech: AI and Medical Robotics, Google Cloud - multiple AI and cloud computing certificates (Generative AI, LLMs, Responsible AI, Vertex AI, Kubernetes Engine, and more)
- **Awards:** VEX Robotics: Design Award for robot design, Congressional Award - Bronze Medal, National Distinguished Student Leadership Award (NDSLA), American Citizenship Award (ACA), American Volunteer Service Award (AVSA) and National Recognition Program award (College Board)

COMMUNITY SERVICE

- Schoolhouse.world - SAT Tutor Jan 2026 - Present
- YAPA Kids - Volunteer Tutor - Dec 2024 - Present
- SKY Meditation Center - Volunteer Tutor - Jan 2024 - Present
- ENGin - Volunteer English Tutor for Ukrainian students - Jul 2024 - Present
- Safe Harbor of Chester County - Volunteer - Jan 2024 - Dec 2024

LANGUAGES

English (Fluent), Tamil (Fluent) and Spanish (Polite)